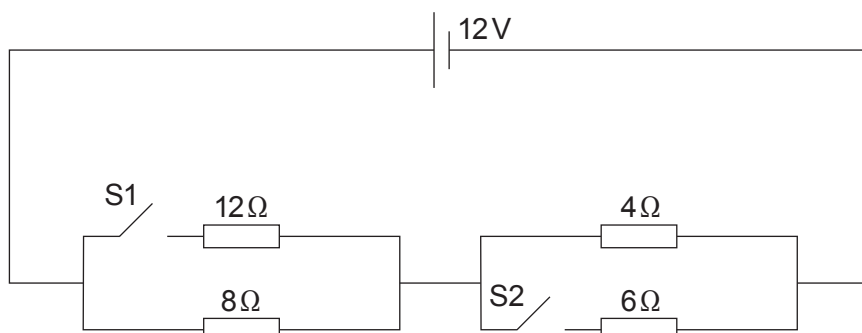


5. Students investigated the following electrical circuits.



Select the appropriate equations to answer the following questions.

$$\text{voltage} = \text{current} \times \text{resistance}$$

$$R = R_1 + R_2$$

$$\text{power} = \text{current}^2 \times \text{resistance}$$

$$\frac{1}{\text{total resistance}} = \frac{1}{\text{total resistance X}} + \frac{1}{\text{total resistance Y}}$$

(a) Calculate the current supplied when both switches are open.

[3]

Current = A

(b) (i) Calculate the total resistance in the circuit.

[3]

Total resistance = Ω

- (ii) Calculate the power dissipated as heat by the 12Ω resistor.

[3]

Examiner
only

Power = W

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